Exercise 2

Database: Employees_db

Table: Employee's

Questions

1. Select Count(Id) As Total number of employees
FROM Employees;

Total number of Employees
10

2. SUM() Function

Select Department, first_name, Last_name, SUM(Salary) AS Total_Salary

FROM Employees

Where Department = 'IT';

Department	Total_Salary
IT	220 000

3. AVG() Function

Select AVG(Salary), ~~Department~~ AS TotalDepartmentSalary, Department

FROM Employees

Where Department = 'HR';

Department	TotalDepartmentSalary
HR	99000 ÷ 2 = 49500

4. MIN() and Max() Functions

Select MIN(Salary) ~~AS~~ LowestSalary

= 48 000

=

Select MAX(Salary) AS HighestSalary

= 62 000

5. GROUP BY Statement:

```
Select SUM(Salary) AS TotalSalary, Department
FROM Employee
GROUP BY Department;
```

Department	TotalSalary
IT	220 000 ÷ 4 = 55000
HR	99 000 ÷ 2 = 49500
FINANCE	119 000 ÷ 2 = 59500
MARKETING	105 000 ÷ 2 = 52 500

6. GROUP BY and COUNT()

```
Select City, Count (Id) AS Employees_per_~~department~~ City
FROM Employees
GROUP BY City;
```

City	Employees_Per_~~Dept~~ City
New York	2
Chicago	3
Los Angeles	2
San Fransisco	2
Houston	1

7. GROUP BY and ORDER BY

```
Select Department, AVG (Salary) AS Average_Salary
FROM Employees
GROUP BY Department
ORDER BY Average_Salary DESC;
```

Department	Average_Salary
~~IT~~ Finance	59500
~~HR~~ IT	55000
~~Finance~~ Marketing	~~49500~~ 52500
~~Marketing~~ HR	49500

8. HAVING CLAUSE

```
Select Department, SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department
HAVING TotalSalary >100 000;
```

Department	TotalSalary
IT	220 000
Finance	119 000
Marketing	105 000

9. Combining GROUP BY, HAVING and ORDER BY

```
Select City, Count(Id) AS Number_of_employees
FROM Employees
GROUP BY City
HAVING Number_of_employees > 1
ORDER BY Number_of_employees DESC;
```

City	Number of Employees
New York	2
Los Angeles	2
Chicago	3
San Francisco	2

10. Combining Aggregate Functions

```
Select Department, Max(AVG(Salary)) AS Highest_AVG_Salary
FROM Employees
GROUP BY Department;
```

Department	Highest_AVG_Salary
Finance	59500